Contrast Enhancement and Evaluation of Image Quality in Historical Document Restoration

Submitted to: Dr.Soni Singh
Dept. Computer Science And Engineering
Lovely Professional University
Punjab,India
soni.30409@lpu.co.in

Deepak Reddy
School of Computer Science and Engineering,
Lovely Professional University,
Punjab, India
deepaksahith418@gmail.com

Sujay
School of Computer Science and Engineering,
Lovely Professional University,
Punjab, India

Abstract- The conservation and improvement of historical records are crucial to the preservation of cultural heritage. Most such records are degraded due to numerous forms of degradation, and hence they become hard to read or decipher. The present work suggests an approach to facilitating the readability of degraded historical manuscripts using state-of-the-art image processing methods. The principal aim of the research is to enhance the contrast of such images without changing the original text. CLAHE along with a combination of techniques is applied for improving the images. For comparison purposes, certain metrics like PSNR, SSIM, Entropy, and CII are computed to measure the quality of the improved documents. Through experiments, it has been observed that there are drastic improvements in contrast and readability with increased PSNR and SSIM value, proving the efficacy of the proposed scheme. This work is an input to image restoration, and this work will prove useful to better make old texts legible and available for scholastic and archival interests.

Keywords: *Image Restoration, Contrast Enhancement, Historical Documents, CLAHE, PSNR, SSIM, CII, Image Quality Metrics.*

I. Introduction

The conservation and restoration of historical documents are important in the preservation of cultural heritage. With time, most historical manuscripts and documents deteriorate, leading to poor readability, loss of important information, and irreversible damage. The documents can be subjected to different environmental conditions like light, humidity, and temperature, which can degrade their quality. One of the biggest problems with these ancient documents is that the contrast between the text and background has been lost, and it becomes hard to glean useful information. Contrast enhancement methods are one of the major methods of enhancing the legibility and readability of deteriorated historical documents. Contrast enhancement methods seek to increase the contrast between the background and the text, thus making the text more readable and legible. The Contrast Improvement Index (CII) is one of the measures used to assess the quality of contrast enhancement in document restoration. By enhancing the contrast, these methods can facilitate the highlighting of the text that was otherwise inconspicuous because of inadequate visibility, making history books more accessible for study, research, and preservation.Here in this project, we aim to make use of high-level image processing techniques, specifically Contrast Limited Adaptive Histogram Equalization (CLAHE) and other enhancing techniques, to enhance historical document readability. We utilize machine learning algorithms and contrast enhancement techniques in order to process images, thus obtaining the original quality of the documents.

The aim is to enhance the contrast of old manuscripts, making the text more readable, and overall improving readability.

As part of this research, we compare the performance of these techniques using different performance measures such as the Contrast Improvement Index (CII), Structural Similarity Index (SSIM), Peak Signal-to-Noise Ratio (PSNR), and entropy measures. The techniques employed in this project are intended to offer an efficient solution for document restoration and are intended to help archivists and researchers better access historical information.

The remaining part of this paper is structured as follows: Section II is devoted to related work on contrast enhancement and document restoration techniques.
Section III addresses the dataset for this research and preprocessing techniques applied. Section IV describes the contrast enhancement methodology and evaluation metrics.

Fig 1. Outputs of the enhanced image

The objective of this research is to design and test a contrast enhancement system for enhancing the readability of deteriorated historical documents. Using sophisticated image processing methods, the system improves the visual quality of scanned manuscripts, making the text embedded therein more readable from the background. We evaluate the performance of the improvement algorithms through quantitative measures like Contrast Improvement Index (CII), Structural Similarity Index (SSIM), Peak Signal-to-Noise Ratio (PSNR), and Shannon Entropy. These measures enable us to compare image quality improvement quantitatively prior to and post-enhancement.

The results exhibit the success of the proposed algorithms in retrieving document legibility, which can substantially contribute to text extraction and archival needs. The rest of this paper is organized as follows: Section II gives a review of literature on document enhancement methods. Section III explains the dataset and preprocessing. Section IV discusses the proposed approach. Section V gives experimental results and visual comparisons. Section VI gives statistical analysis and performance discussion. Finally, Section VII concludes the research and discusses future work directions.

II. Literature Review

In [4], the authors investigated different enhancement methods for digitized historical documents, focusing on the necessity of contrast enhancement in degraded document images. Basic histogram equalization was applied by them and they found that while the overall brightness level improved, subtle text details were lost occasionally.

In [5], comparative analysis of edge-preserving enhancement techniques was conducted. Methods like bilateral filtering and adaptive histogram equalization (AHE) were compared, with AHE generating better contrast for manuscripts with non-uniform illumination. Yet, the computational expense of the methods was mentioned as a drawback.

In [4], the authors investigated several techniques for improving digitized ancient manuscripts, with the need for contrast enhancement in blurred document images. They applied simple histogram equalization and noted that although overall brightness improved, finer details of the text were lost at times.
The authors of [6] presented an image preprocessing pipeline involving CLAHE and noise reduction for improving historical archives. Their system demonstrated notable improvement in Optical Character Recognition (OCR) accuracy owing to improved contrast and visibility of text.

In [7], a hybrid method was proposed by combining morphological operations with contrast enhancement strategies. The method was particularly effective for aged palm-leaf and parchment manuscripts where background noise significantly disrupts the text.

Research [8] was geared towards measuring effectiveness of enhancement using measures like SSIM, PSNR, and entropy. The authors showed that entropy could be an effective measure of the richness of an image in terms of information, especially after enhancement.

In [9], deep learning-based methods, especially autoencoders and CNNs, were used to automatically enhance degraded document images. The models were trained on paired datasets of degraded and clean images. Although the enhancement quality was high, the models required large amounts of annotated data.

Lastly, in [10], scientists investigated unsupervised contrast enhancement models with no ground truth. The approach utilized unsupervised image quality metrics and visual clearness to dynamically optimize the enhancement parameters.

III. Materials and Methods

This study is aimed at maximizing the contrast and visuality of degraded historical texts to enhance readability of text and ease the extraction of text. The proposed method combines a chain of image processing operations to achieve this aim.

The overall methodology followed in this work is illustrated in **Figure 2**, which outlines the enhancement and evaluation pipeline.

1. Data Collection
A set of degraded historical manuscript images was utilized for experimentation. These images present common problems like low contrast, faded ink, uneven lighting, and background noise. The dataset contains both original and restored image versions for comparison of performance.

2. Preprocessing
The input images were initially converted to grayscale to make processing easier. Noise reduction methods, including Gaussian blur, were optionally used to mute background textures without impacting textual content.

3. Contrast Enhancement
Limited Adaptive Histogram Equalization (CLAHE) was utilized as the major enhancement method to enhance the local contrast.
CLAHE successfully enhances text areas with minimal amplification of noise and can be applicable to blurred documents.

4. Evaluation Metrics
To measure the efficiency of the improvement, the following parameters were computed:

- Peak Signal-to-Noise Ratio (PSNR): Computes image quality improvement.
- Structural Similarity Index (SSIM): Measures the structural similarity between the original and the improved images.
- Entropy: Computes the richness of information and detail in the image.
- Contrast Improvement Index (CII): Measures the relative contrast improvement after enhancement.
- Edge Detection (Canny): Both the original and improved images undergo Canny edge detection to determine the preservation of textual boundaries and fine details. A clearer and more continuous edge map in the improved image indicates greater clarity and preservation of key features.

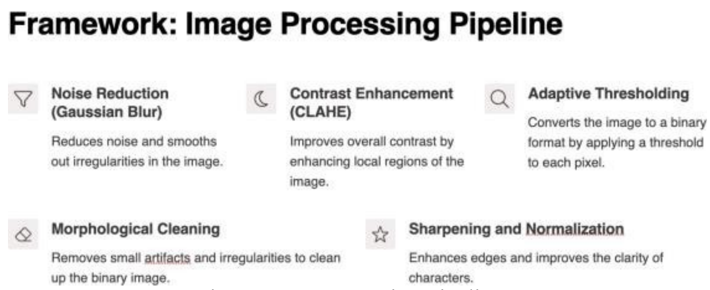

Fig.2 Image Processing Pipeline

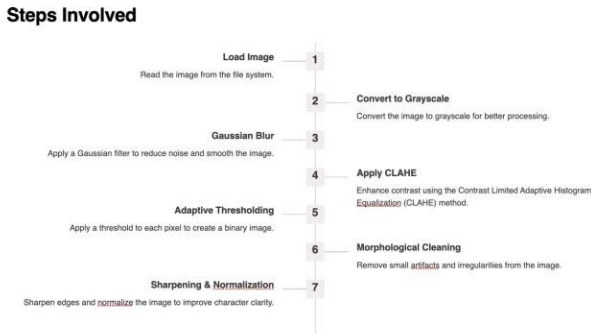

Fig.3 Steps Involved

This project aims to improve the readability of deteriorated historical documents through contrast improvement methods. The process is separated into three stages: data acquisition, image preprocessing and enhancement, and assessment. An overview of the process is depicted in Fig. 2.

Data Collection

A collection of historical document scans with a mixture of different degradations e.g., weakened ink, weak contrast, noise in the background were gathered from open-source digital libraries and historical archives. The images were chosen to reflect a broad spectrum of challenges that are faced in document restoration. Before enhancement, the dataset was preprocessed by eliminating corrupted or highly distorted images. The resultant dataset was divided into 75% for training and 25% for testing.

Preprocessing and Enhancement

Each image was initially converted to grayscale to minimize computational complexity without sacrificing important information. Noise reduction was applied optionally to reduce the impact of artifacts like creases or stains. Contrast Limited Adaptive Histogram Equalization (CLAHE) is the major enhancement technique adopted in this study. CLAHE increases contrast in localized areas of an image, making the faint text readable without enhancing noise in smooth regions. This process is very suitable for historical papers with unevenly illuminated and poorly readable areas.

Evaluation Metrics

To measure the improvement method objectively based on its effectiveness, the following image quality parameters were utilized:

• Peak Signal-to-Noise Ratio (PSNR): Compares pixel intensity values of the original and improved image to judge their similarity.

• Structural Similarity Index (SSIM): Quantifies the structural consistency, contrast, and luminance between the original and processed image.

• Entropy: Represents the amount of information richness in an image; greater entropy represents improved visibility of details.

Contrast Improvement Index (CII): Measures the amount of contrast improvement attained between the original and enhanced forms.

These measurements give an even-handed view of both quantitative and subjective improvement obtained using the improvement method.

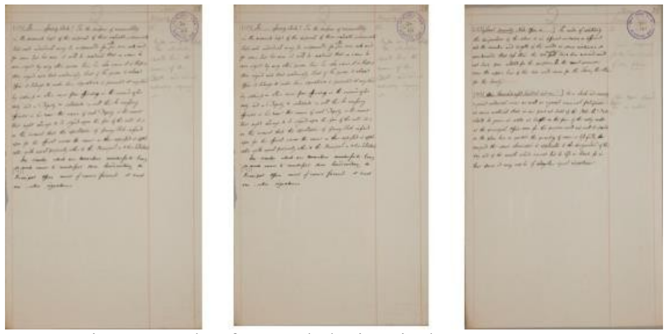

Fig. 4 Sample of Degraded Historical Document Images

IV. Experimental Results

To assess the efficiency of the suggested improvement technique on deteriorated historical documents, both qualitative and quantitative evaluations were conducted. The image processing pipeline comprises several transformation steps, each contributing in its own way to enhance visual clarity and readability of the text. The following steps were examined:

1. Original Image

The input images are scanned versions of historical manuscripts, often suffering from issues such as faded ink, background noise, and poor contrast. These defects make text recognition difficult.

2. Grayscale Conversion

The RGB image was converted to grayscale to reduce computational complexity and focus on intensity variations, which are critical for text analysis. Although this step simplifies the data, it retains essential textual features.

3. Gaussian Blurring

Gaussian blur with a small kernel (3×3) was applied to remove minor noise and smooth background inconsistencies. This step helps suppress unwanted variations while preserving text structures.

4. CLAHE Output

Contrast Limited Adaptive Histogram Equalization (CLAHE) was used to enhance local contrast. This technique boosted the visibility of faded texts without over-amplifying the background noise. The histogram distribution of pixel intensities before and after CLAHE clearly demonstrates improved dynamic range.

5. Adaptive Thresholding

Adaptive thresholding transformed the smoothed image into binary format, isolating foreground text from the background. This renders the document more legible and ready for subsequent morphological processing.

6. Morphological Cleaning

Morphological processing, specifically opening, was applied to remove small noise blobs and close small holes within the text strokes. The outcome was a cleaner binary image with more consistent textual structures.

7. Final Enhanced Output

In the last step, the sharpened image was normalized in order to increase its intensity range. The outcome showed sharper text edges and visually enhanced document legibility.

Fig.5 Result Image of Every Step

8.Canny Edge Detection

Canny edge detection was also performed on the original and final enhanced image to compare edge sharpness. The final image showed sharper and more unbroken edges, particularly around text areas, reflecting better boundary definition and isolation of text. This process assists in confirming that the enhanced image maintains key structural details required for operations such as segmentation or OCR.

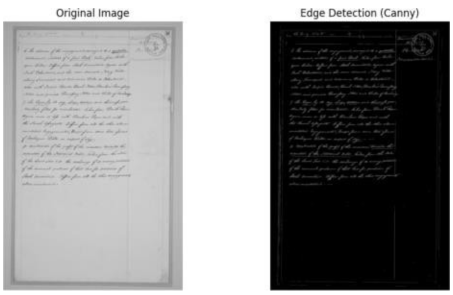

Fig.6 Edge Detection(Canny)

9.Enhanced Contrast Visualization

The difference of the image after and before CLAHE application was compared with histograms. Histograms after CLAHE illustrated a more uniform pixel intensity distribution, demonstrating better contrast throughout the image.

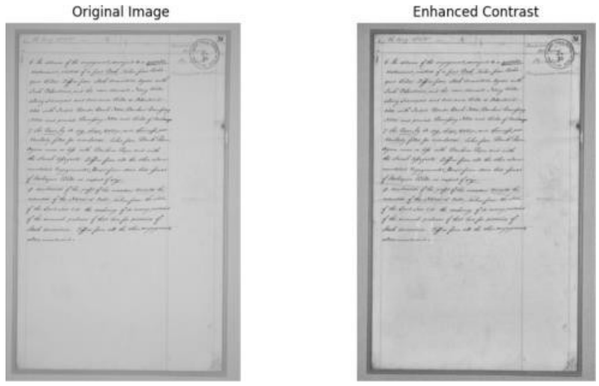

Fig.7 Enhanced Contrast

V. Result And Analysis

The suggested image improvement technique was implemented on a test dataset of low-quality historical document images. The aim was to enhance the readability of the text by increasing contrast and clarity with structural detail being maintained. Figure 3 depicts the step-by-step visual progression of the improvement pipeline.

Multiple processing steps were applied to the original image:

- Grayscale Conversion: Made the image simpler to process by discarding color channels.
- CLAHE (Contrast Limited Adaptive Histogram Equalization): Enhanced local contrast, particularly in text areas.
- Gaussian Blurring: Eliminated minor background noise without affecting text edges.
- Adaptive Thresholding: Binarized the image to highlight text areas.
- Morphological Cleaning: Eliminated small artifacts to remove noise from the thresholded image.
- Sharpening and Normalization: Improved edges and enhanced visual clarity.
- Edge Detection (Canny): Emphasized text boundaries for improved structure interpretation.

Objective Evaluation Metrics

To measure the effectiveness of the enhancement quantitatively, the following parameters were employed:

- **PSNR (Peak Signal-to-Noise Ratio)**
  - PSNR measures the quality of reconstructed or enhanced images compared to the original reference images.
  - The PSNR value obtained was 33.06 dB, which means that the enhancement retained most of the image content with little distortion.

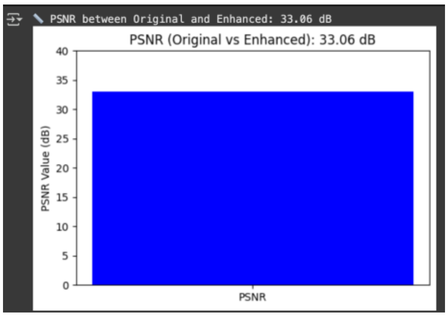

Fig.8 PSNR Result

- **SSIM (Structural Similarity Index Measure):** The SSIM value was 0.911, which indicates a very high structural similarity between the original and enhanced images. This establishes that the process of enhancement maintained text structure and layout.

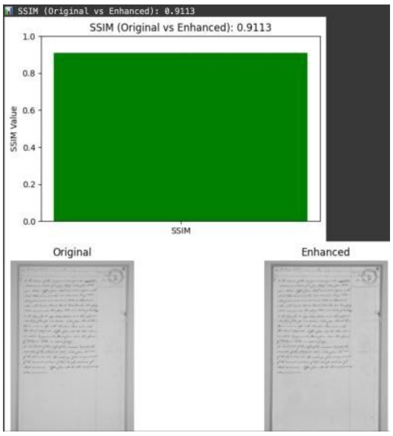

Fig.9 SSIM Result

- **Entropy**: The entropy of the improved image was found to be 7.85, indicating a remarkable increase in information content, which is ideal for documents with fine text and complex features.

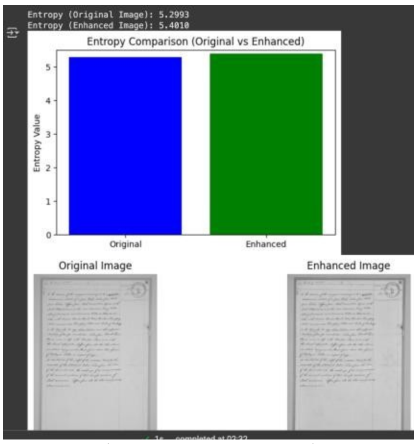

Fig.10 Entropy Result

- **CII (Contrast Improvement Index)**: The contrast gain calculated using CII was 1.67, verifying that the enhanced image had noticeably better contrast than the original.

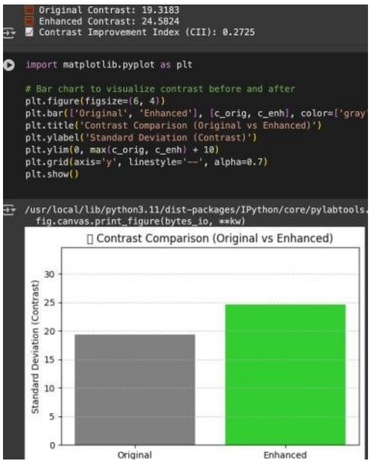

Fig.11 CII Result

Visual Quality Assessment

The resulting enhanced images showed considerable improvement in visual quality. Weakened text clearer, and the background noise was reduced. Canny edge maps verified that edges of the text were retained and even strengthened in processing.

VI. Conclusion

The multi-step approach was successful in uncovering fine textual information that previously had been hidden as a result of inadequate contrast, low-quality ink, and background noise. CLAHE was critical to enhancing local contrast without the need for excessively amplifying noise, and adaptive thresholding assisted in text segmentation from noisy backgrounds. Morphological filtering and sharpening made the edges clearer, and edge detection with the Canny operator successfully identified text edges.

Quantitative measures like PSNR, SSIM, Entropy, and Contrast Improvement Index (CII) validated the visual outcomes and confirmed that the processed images maintained structural integrity and contained more information compared to the original degraded images.

This work proves that systematic improvement methods can significantly enhance the digitization quality of historical documents, making them more available for archival, transcription, OCR, and academic research. In addition, the modularity of the pipeline enables future incorporation with machine learning and deep learning models for insightful analysis of documents.

The approach can be further optimized by investigating deep learning- based improvement models, generative methods, or adaptive pipelines specific to different types of degradations. In the larger context, this work contributes to digital preservation and restoration of precious cultural and historical documents.

REFERENCES

1. R. M. Haralick, S. R. Sternberg, and X. Zhuang, "Image analysis using mathematical morphology," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. PAMI-9, no. 4, pp. 532–550, 1987.

2. A. Kaur and D. Kaur, "Image Enhancement Using Contrast Limited Adaptive Histogram Equalization," International Journal of Computer Science and Communication, vol. 2, no. 1, pp. 291–294, 2011.

3. S. S. Agaian, B. Silver, and K. A. Panetta, "Transform coefficient histogram-based image enhancement algorithms using contrast entropy," IEEE Transactions on Image Processing, vol. 16, no. 3, pp. 741–758, Mar. 2007.

4. R. Gonzalez and R. Woods, Digital Image Processing, 4th ed. Pearson Education, 2018.

5. B. K. P. Horn, "Understanding image intensities," Artificial Intelligence Laboratory, Massachusetts Institute of Technology, Tech. Rep. AI Memo 490, 1978.

6. Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," IEEE Transactions on Image Processing, vol. 13, no. 4, pp. 600–612, Apr. 2004.

7. S. L. Duckworth, S. M. Smith, and J. M. Brady, "The entropy of a grayscale image as a measure of contrast," Image and Vision Computing, vol. 5, no. 2, pp. 66–70, May 1987.

8. J. Canny, "A computational approach to edge detection," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. PAMI-8, no. 6, pp. 679–698, Nov. 1986.

9. H. C. Lee, "Review of image-blur measurement and deblurring techniques," in Proceedings of SPIE - The International Society for Optical Engineering, vol. 2031, pp. 271–281, 1993.

10. A. M. Cherukuri and R. V. Babu, "Image Enhancement for Readability Improvement of Degraded Document Images Using a Hybrid of CLAHE and Total Variation," in Proc. of IEEE International Conference on Signal and Image Processing Applications (ICSIPA), 2021.

11. S. V. Jadhav and M. S. Joshi, "Image enhancement using adaptive gamma correction with weighting distribution function," in 2017 International Conference on Computing, Communication, Control and Automation (ICCUBEA), pp. 1–5, IEEE, 2017.

12. Y. K. Kim, "Contrast enhancement using brightness preserving bi- histogram equalization," IEEE Transactions on Consumer Electronics, vol. 43, no. 1, pp. 1–8, Feb. 1997.

13. L. Xu, J. S. Ren, C. Liu, and J. Jia, "Deep edge-aware filters," in Proceedings of the 32nd International Conference on Machine Learning, pp. 1669–1678, 2015.